ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

TECHNOLOGY

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 75

Marking guideline for the Academic Central Grade 7 Technology paper 1 · practice d.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

QUESTION 1

1 1.1 B ✓ 1.2 C ✓ 1.3 D ✓ 1.4 C ✓ 1.5 A ✓ 1.6 B ✓ 1.7 D ✓ 1.8 C ✓ 1.9 A ✓ 1.10 B ✓ [10]

QUESTION 2

2.1 Investigate ✓, design ✓, make, evaluate ✓, communicate (3 marks for the correct order; deduct 1 for each step out of place) (3)

2.2.1 Any sensible brief ✓✓, e.g. "Design and make a strong, stable tower that holds a 1 000-litre water tank 4 m above the ground so that water can flow to the village taps." (2)

2.2.2 Four legs / columns ✓; platform for the tank at the top ✓; diagonal cross-bracing forming triangles ✓; neat sketch with labels ✓ (4)

2.2.3 Suitable material ✓ with a reason ✓, e.g. steel – strong and can carry the heavy load (galvanise or paint it against rust); treated poles – cheap and strong (2)

2.2.4 Any TWO ✓✓: a wide base (legs spread out); anchor the legs in concrete foundations; cross-bracing; keep the structure symmetrical; guy ropes or stays (2)

2.3 The point where all the weight of an object seems to act ✓ (the balance point) ✓ (2)

2.4 A frame structure is made of parts (members) joined together to form a skeleton, e.g. a pylon or a roof truss ✓; a shell structure has a thin outer covering that carries the load and protects the inside, e.g. an egg, a helmet, a can ✓; one correct example of each ✓ (3)

[18]

QUESTION 3

3.1 Difference ✓, e.g. hardwood comes from broad-leaved (deciduous) trees that grow slowly and is usually harder and heavier; softwood comes from conifers that grow fast. Hardwood example ✓: oak, teak, meranti, kiaat, yellowwood. Softwood example ✓: pine, spruce, cedar. (3)

3.2 A temporary joint can be taken apart without damaging the parts ✓, e.g. nuts and bolts, screws ✓; a permanent joint cannot be taken apart without damage ✓, e.g. welding, gluing, rivets, nails ✓. (4)

3.3 (a) Tenon saw (or panel saw) ✓ (b) Tin snips ✓ (c) Sandpaper (or a file / plane) ✓ (3)

3.4 Any TWO ✓✓: use the right tool for the job; keep tools sharp and in good condition; clamp the work in a vice; cut away from your body; wear safety glasses; carry tools safely and pack them away (2)

[12]

QUESTION 4

4.1 (a) First class ✓ (b) Second class ✓ (c) Third class ✓ (3)

4.2.1 Bar resting on a fulcrum ✓; rock (load) on the short end ✓; effort (arrow) on the long end ✓ (3)

4.2.2 $600 \times 0{,}5 = \text{effort} \times 1{,}5$ ✓; $300 = \text{effort} \times 1{,}5$ ✓; $\text{effort} = 200\ \text{N}$ ✓ (3)

4.2.3 $\text{MA} = 600 \div 200$ ✓ $= 3$ ✓ (2)

4.2.4 Use a longer crowbar / push further from the fulcrum, or move the fulcrum closer to the load ✓ (1)

4.3 It makes the output move in the opposite direction to the input ✓; any example ✓, e.g. pull-type toys, windscreen wipers, bicycle brake mechanisms (2)

4.4 It makes the output move in the same direction as (parallel to) the input ✓; any example ✓, e.g. a toolbox with cantilever trays, a desk lamp, scissor lift (2)

4.5 The effort (the biceps muscle) is between the fulcrum (the elbow) and the load (in the hand) ✓✓ (2)

[18]

QUESTION 5

5.1 Correct symbols ✓✓✓✓: (a) long thin line and short thick line; (b) circle with a cross; (c) a break with the switch lever lifted; (d) a rectangle (or zigzag) (4)

5.2 Two cells in series correctly drawn ✓; switch in the main circuit ✓; two bulbs on separate parallel branches ✓; closed, neat circuit with correct symbols ✓ (4)

5.3 (a) Conductor ✓ (b) Insulator ✓ (c) Conductor ✓ (d) Insulator ✓ (4)

5.4 Any TWO ✓✓: more turns of wire; more cells (bigger current); a thicker iron core (2)

5.5 Use ✓, e.g. a crane at a scrapyard; an electric bell; a door lock; a relay. Reason ✓: it can be switched on and off, so the load can be picked up and dropped (or the strength can be changed). (2)

5.6 Electrical energy → kinetic (movement) energy (plus some sound and heat) ✓ (1)

[17]

TOTAL: 75

END OF MARKING GUIDELINE